
Feedback That Backfires: Why Small Language Model Agents Repeat the Call They Just Watched Fail

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Abstract

1 Agent harnesses record a failed tool call and its error message in the transcript
2 and ask the model to continue, on the reasonable assumption that the error is cor-
3 rective information. We measure whether it is. Defining the *corrective gain* of a
4 failure record as the change in log-probability of re-emitting the action that just
5 failed, we find the gain is *negative* for every instruction-tuned model we tested
6 (6 checkpoints, 135M–1.7B, 4 families) in two environments: simulated tool call-
7 ing and MBPP program repair with real interpreter output. Normalised by action
8 length the effect has the same size in both, about -1.03 nats per action token, a
9 factor of 2.8 in the odds of each token. It holds on 90%–100% of individual items,
10 not merely on average. Over a fixed candidate set the normalised probability of
11 repeating the failed call rises from 0.06 to 0.54, and greedy decoding reproduces
12 the failed call token for token on 19% of items after the failure versus 0% before
13 it. Counterfactual observations, in which the same call is paired with a failure
14 message, a success message, or a valence-free acknowledgement, separate two
15 competing effects. The failed call’s *surface form* accounts for 83% of the damage;
16 the semantic contribution of marking it as failed is small and its sign is not even
17 consistent across environments. The problem is therefore in the harness, not in
18 the model’s grasp of error messages, and that predicts which remedies can work.
19 Replacing the verbatim call with a runtime-generated description of the failure re-
20 moves 76% of the inversion at no token cost, and making previously-failed strings
21 unreachable at the decoder acts on the same term. Two remedies that sound right
22 do not act on it at all. An explicit “do not repeat” instruction leaves the measured
23 quantity where it was; and deleting the failed attempt to retry from a clean con-
24 text, the standard prescription for context contamination, is the worst harness we
25 measured for repetition, because it restores precisely the context that produced the
26 failure. What survives is narrow enough to be useful: the context after a failure
27 must differ from the context before it, and the difference must not be the failed
28 action. The study runs end to end on a CPU and all artefacts are released.

1 Introduction

30 An agent that calls tools will get some of those calls wrong. What happens next is decided by the
31 harness rather than the model: it appends the failed call to the transcript, appends whatever the
32 runtime printed, and asks the model to continue. ReAct [Yao et al., 2022] introduced this pattern
33 and essentially every agent framework since has kept it, including those built for small open-weight
34 models. The premise is that the error message is corrective information, produced by ground truth
35 rather than by a critic.

We measure whether it is, and it is not. Fix a decision point: a context C , an action $a^\times$ that fails there, and the observation $o^\times$ the runtime returns. Define the *corrective gain* as the change in the log-probability of writing $a^\times$ again once the failure has been recorded. Across 6 instruction-tuned checkpoints (135M–1.7B, 4 families) and two environments, the gain is *negative*: recording the failure makes the failed call *more* likely, by -17.38 nats, or about -1.03 nats per action token. The effect is not an average over a few dramatic items; it holds on 90% or more of individual items.

The reason matters more than the size, because it decides which fixes can work. Appending $(a^\times, o^\times)$ does two things at once: it places the failed call’s token sequence where the model’s copying machinery can reach it [Olsson et al., 2022, Xu et al., 2022], and it asserts that the call failed. Scoring the same action against a *valence-free* observation, one that acknowledges the call without saying how it went, separates them. The surface-form term accounts for 83% of the effect. The semantic term, which would have to be large and negative for “the model does not understand the error” to be the explanation, is small and its sign is not even consistent across our two environments.

That localises the problem in the harness rather than in the model, and it predicts which remedies act on the right term. Two do. Replacing the verbatim call with a description the runtime generates from its own error metadata removes 76% of the inversion at no token cost, and banning previously-failed strings at the decoder acts on the same term. Two do not. A natural-language prohibition moves the measured quantity slightly the *wrong* way, as the negative-instruction literature would predict for a prohibition that must name the string it forbids [Castricato et al., 2024, Vrabcová et al., 2025]. More surprisingly, deleting the failed attempt and retrying from a clean context, which is what prior work on context contamination recommends [Yang, 2026], is the *worst* harness we measure: it multiplies the exact repeat rate by 2.6 without improving task success. Deleting the failure restores the exact context that produced it, and a deterministic policy in an identical context emits an identical action.

What survives is narrow enough to be useful:

The context after a failure must differ from the context before it, and the difference must not be the failed action itself.

Contributions. (i) A measurement, the corrective gain of a failure record, shown negative across 6 models and two environments, reported alongside a normalised repeat probability and an exact greedy repeat rate. (ii) A decomposition into a surface-form and a semantic term, which identifies the surface form as the cause and predicts which interventions can work. (iii) An evaluation of three harness interventions in free-running rollouts, including one that follows from prior work and makes matters worse. Everything runs on a CPU; code, item sets and raw per-score records are released.

2 Corrective gain and its decomposition

Write $\oplus$ for appending a turn pair to the transcript, so that $C_{t+1} = C_t \oplus (a_t, o_t)$. Almost every agent framework implements that literally, which is the design decision this paper is about. For a decision point $(C, a^\times, o^\times)$ define

$$G(a^\times) = \log \pi(a^\times | C) - \log \pi(a^\times | C \oplus (a^\times, o^\times)), \quad (1)$$

where $\log \pi(a | c) = \sum_i \log \pi(a_i | c, a_{<i})$. Both terms score *the same string*, so the difference is exactly the change in the log-odds of writing that action again, and it is invariant to the action’s length and to how the tokeniser splits it. $G > 0$ is the behaviour the harness is designed to produce. $G < 0$ we call *feedback inversion*.

Let $o^\emptyset$ be a *valence-free* observation, which acknowledges the call but reports neither success nor failure. Then with $\Delta_{\text{copy}} = \log \pi(a^\times | C \oplus (a^\times, o^\emptyset)) - \log \pi(a^\times | C)$ and $\Delta_{\text{sem}} = \log \pi(a^\times | C \oplus (a^\times, o^\times)) - \log \pi(a^\times | C \oplus (a^\times, o^\emptyset))$,

$$-G(a^\times) = \Delta_{\text{copy}} + \Delta_{\text{sem}} \quad (2)$$

by construction. Δ_{copy} isolates what merely *having written the action* does, holding the feedback content neutral; Δ_{sem} isolates what *marking it as failed* adds, holding the surface form fixed.

Equation 2 makes the design space concrete. If Δ_{sem} dominates, the model misread the error and the remedy is a better model or a clearer message. If Δ_{copy} dominates, message engineering cannot

Model	Params	G	G/token	copy	sem	G_{abs}
SmolLM2-135M	0.14B	-23.92 [-25.86, -22.09]	-1.21 [-1.36, -1.06]	21.69 [19.54, 23.94]	2.23 [1.59, 2.90]	-2.24 [-3.41, -1.05]
SmolLM2-360M	0.36B	-18.62 [-21.00, -16.38]	-0.93 [-1.06, -0.79]	16.08 [13.89, 18.43]	2.54 [1.98, 3.16]	-3.23 [-4.24, -2.24]
Qwen2.5-0.5B	0.49B	-13.91 [-16.12, -11.82]	-0.94 [-1.14, -0.75]	9.91 [7.59, 12.25]	4.00 [2.82, 5.28]	-2.86 [-3.71, -1.99]
Qwen2.5-0.6B	0.60B	-21.50 [-25.07, -18.18]	-1.47 [-1.74, -1.21]	18.25 [14.75, 21.95]	2.78 [1.56, 3.96]	-6.60 [-8.07, -5.25]
Llama-3.2-1B	1.24B	-12.41 [-15.71, -9.62]	-0.92 [-1.21, -0.66]	9.42 [7.23, 11.91]	3.26 [2.06, 4.62]	-5.46 [-7.49, -3.65]
SmolLM2-1.7B	1.71B	-13.91 [-15.53, -12.35]	-0.71 [-0.81, -0.61]	11.67 [10.37, 12.97]	2.24 [1.44, 3.12]	-4.36 [-5.08, -3.57]

Table 1: Probe results on ToolShed (tool calls). All quantities are in nats and are differences in the summed log-probability of the *same* action string under two contexts. $G > 0$ means the harness’s failure record made the failed call less likely; $G < 0$ is feedback inversion. Cells give the mean over items with a 95% cluster bootstrap interval (clustered on task); bold marks intervals excluding zero.

reach the problem, because the damage is done by the presence of the string rather than by anything said about it, and the remedies must be structural: remove the surface form from the context, or make it unreachable at decoding time.

3 Setup

Environments. *ToolShed* is a simulated office workspace with twelve typed tools and six task templates requiring two to four calls. Failing actions are constructed from the reference action by 8 perturbation operators yielding 12 error families, so every model is probed on byte-identical items and a scaling curve cannot confound model size with item difficulty. We built it because the decomposition needs, for one fixed action, an observation reporting failure, one reporting success and one reporting nothing; standard agent benchmarks are built to score policies and do not expose counterfactual observations. *CodeRepair* is the sanitised split of MBPP [Austin et al., 2021], executed in a subprocess against the problem’s own assertions, where failing programs are single-edit mutations and nothing about the error text is authored by us. It also exercises a regime *ToolShed* cannot: actions are whole functions, and a real traceback quotes the offending source line back at the model.

Models and protocol. 6 instruction-tuned checkpoints spanning 135M–1.7B across 4 families (Llama-3.2, Qwen2.5, Qwen3, SmolLM2), in float32 on a CPU with 17 GB of RAM, which is what sets the parameter ceiling. Scoring is teacher-forced and deterministic. Items from one task share a prompt and a prefix, so every interval is a 95% cluster bootstrap over *tasks* (10,000 resamples); the headline family is Holm-corrected across models. A model/environment cell with fewer than 30 scored items is excluded from every table and fit.

4 Execution feedback inverts, and the string is why

Table 1 reports the gain and its decomposition on *ToolShed*. It is negative for every model, and the intervals exclude zero for 6 of 6. Two readouts make the size concrete without reference to nats. Over a fixed four-action candidate set the normalised probability of re-emitting the failed call rises from 0.06 to 0.54; and greedy decoding, which is what the agent would actually have written next, reproduces the failed call token for token on 19% of items after the failure against 0% before it.

The decomposition is the point, and Figure 1 is the whole argument in one panel. Δ_{copy} accounts for 83% of the total. Δ_{sem} is small, and on *ToolShed* it is slightly *positive*, plausibly because explaining what went wrong requires naming the tool and the offending argument again; on *CodeRepair* it is slightly negative, so a traceback does push the model away from the program that produced it. We stress the second case because it fixes the interpretation: these models are not simply failing to read error messages. Whatever reading happens is swamped, by a median of 7 to one and by 106 to one at the extreme.

Copying does not simply raise everything. The correct action shares most of its tokens with the failed one, so its probability rises too, by 4.30 nats. What governs the next move is the margin, and the margin moves the wrong way: the gold-minus-failed gap shrinks by -13.12 nats once the failure is recorded. The normalised repeat probability is computed over a fixed candidate set, so a uniform lift cancels; it still rises sharply. And the greedy readout is a statement about the argmax, immune to uniform lifts by construction.

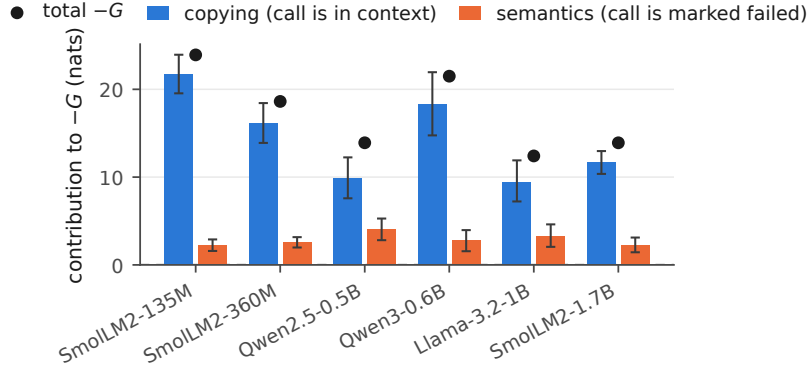


Figure 1: Where the inversion comes from. Bars decompose $-G$ into the effect of the failed call being present in the context (blue) and the additional effect of marking it as failed (orange); black dots are the total. Error bars are 95% cluster bootstrap intervals. If the problem were that the model does not understand the error, the orange bars would be large and negative.

Harness variant	SmolLM2-135M	Qwen2.5-0.5B
terse error text	-0.32 [-0.72, 0.02]	-0.23 [-0.64, 0.20]
verbose error text	-0.09 [-0.32, 0.14]	-0.31 [-0.65, 0.02]
error quotes the failed call	1.31 [1.12, 1.51]	0.10 [-0.22, 0.44]
failure placed before the successful steps	-0.94 [-1.45, -0.48]	-0.82 [-1.36, -0.26]
+ "do not repeat" instruction	0.16 [0.06, 0.27]	-0.02 [-0.20, 0.15]
the same call failed twice	3.06 [2.78, 3.36]	3.61 [3.12, 4.08]
the same call failed three times	3.29 [3.01, 3.59]	4.78 [4.18, 5.37]
placebo: a <i>different</i> call failed	-15.71 [-17.38, -14.23]	-11.71 [-13.28, -10.18]
abstracted failure (no verbatim call)	-21.68 [-23.20, -20.14]	-11.05 [-12.80, -9.39]
bare marker, no diagnosis	-24.23 [-26.30, -22.32]	-13.01 [-15.04, -11.10]

Table 2: Harness variants, as *paired* changes in the log-probability of re-emitting the failed call, relative to the standard harness on the same items. Positive means the variant makes repetition *more* likely; negative means less. Every row uses the same items and the same failed call, and only how the failure is written into the transcript changes.

Scale. Within a family, where training data is held roughly fixed, the larger checkpoint has the milder gain, in both environments independently, and the weakening is driven by Δ_{copy} rather than Δ_{sem} : larger models copy less, they do not read the error better. A log-linear fit has slope 9.51 nats per decade ($R^2 = 0.64$) and reaches zero at 37.6B, but we report that number in order to argue against it: refitting with each model held out moves the crossing from 20B to 79B, a factor of 3.8 decided by which six checkpoints were affordable. Whether the effect survives at 8B is the first question a reader with GPUs should ask, and the probe costs a few hundred forward passes per model to answer.

5 What moves the effect

If the attribution to surface form is right, the effect should respond to manipulations of the *form* and be largely indifferent to manipulations of the *message*. Table 2 reports each variant as a *paired* change against the standard harness on the same items.

Message content is nearly inert. Shortening the error to a bare category name changes the quantity by $-0.28 [-0.68, +0.11]$ nats and expanding it with the tool signature and a corrective hint by $-0.20 [-0.48, +0.08]$; both intervals contain zero. Occurrence count is not inert. When the runtime echoes the offending call inside the error message, as a Python traceback does by default, repetition becomes $+0.71 [+0.45, +0.97]$ nats more likely. Adding information changes nothing; adding a second copy of the string changes something.

Removing the surface form does the work. Replacing the call with a runtime-generated description moves the quantity by $-13.22 [-15.30, -11.34]$ nats and drives the exact greedy repeat rate to zero for every model. Stripping our wording from that description, so the transcript reads only [attempt 1 failed], recovers $-18.62 [-20.67, -16.71]$ against $-16.37 [-18.00, -14.76]$ for the full abstraction on the same models, so the benefit is the absence of the string rather than our glosses. A placebo in which

Harness	success	Qwen2.5-0.5B	
		exact rep.	canon. rep.
verbatim (standard)	0.42 [0.21, 0.62]	0.31 [0.15, 0.47]	0.31 [0.15, 0.47]
+ “do not repeat”	0.58 [0.38, 0.79]	0.24 [0.08, 0.42]	0.24 [0.08, 0.42]
abstract	0.33 [0.17, 0.54]	0.16 [0.04, 0.31]	0.16 [0.04, 0.31]
drop	0.33 [0.17, 0.54]	0.80 [0.79, 0.81]	0.80 [0.79, 0.81]
verbatim + ban	0.42 [0.21, 0.62]	0.08 [0.00, 0.15]	0.08 [0.00, 0.15]
abstract + ban	0.33 [0.17, 0.54]	0.07 [0.00, 0.14]	0.07 [0.00, 0.14]

Table 3: End-to-end rollouts on ToolShed. Success is the environment’s goal predicate; *exact rep.* is the fraction of failed actions that repeat an earlier failed action byte for byte, and *canon. rep.* applies the same count after parsing and normalising the call, so paraphrases of a banned string are still counted. Brackets give 95% cluster bootstrap intervals over tasks.

145 a *different* plausible-but-wrong call is recorded as having failed recovers -13.71 $[-15.33, -12.20]$ nats, a
146 large majority but not all of the effect; since the distractor shares most of its tokens with the scored
147 action, it leaks copying by construction and bounds the string-specific component from below.

148 6 Free-running rollouts

149 The probe holds the trajectory fixed. We also let a model act freely in ToolShed over 24 tasks for up
150 to 6 steps with greedy decoding, comparing 6 harnesses that differ only in what the model sees and
151 how its next token is chosen.

152 Under the standard harness the agent solves 42% of tasks and 31% of its failed actions are byte-
153 for-byte repeats. The prohibition does not reduce repetition (-17 $[-44, +12]$ points, interval containing
154 zero), agreeing with the probe. Clearing the context is the worst case: exact repetition rises to
155 80%, a paired difference of $+49$ $[+33, +64]$ points, and task success does not improve. Both surface-
156 form interventions reduce repetition, the decoder ban most cleanly, taking the exact rate to 8%
157 (-23 $[-38, -10]$ points) and loops from 29% of rollouts to 12%, at no cost in generated tokens.

158 Ranked by loop rate the harnesses order exactly as the decomposition predicts: the two interventions
159 acting on the failed call’s surface form at the good end, the harness that deletes the failure at the bad
160 end, the standard harness between. We report two things that cut against us. Task success does
161 not follow that ordering, because at this scale repetition is a substantial but not majority share of
162 what goes wrong: 18% of actions are prose where a call was expected, which no arrangement of
163 the transcript repairs. And the ban displaces repetition into paraphrase, which is why we report a
164 canonical repeat rate alongside the exact one and do not recommend the ban alone.

165 7 Limitations and conclusion

166 The ladder stops at 1.7B because of a memory limit, not a scientific choice, and the extrapolated
167 crossing is unstable under leave-one-out, so we characterise the small-model regime and a direction
168 of travel rather than a threshold. The rollout study is one model on 24 tasks; the intervals are cor-
169 respondingly wide. Failing actions are constructed rather than sampled, which buys byte-identical
170 items across the ladder at the cost of some realism, and the rollout study, where models generate
171 their own failures, is where that trade is tested. ToolShed is ours; CodeRepair, whose observations
172 are written by the Python interpreter, is the check on it and shows the same pattern.

173 The claim we make is narrow. Execution feedback is not useless and agent transcripts should not be
174 thrown away. But the standard way of *delivering* that feedback hands a small model a strong reason
175 to repeat itself and a weak reason not to, and a harness can keep the diagnosis while withholding the
176 reason. For agents built on small models, which is where these effects are largest and where cost
177 pressure is greatest, that is a change to the harness and not to the model.

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Model	Params	G	G/token	copy	sem	G_{abs}
SmolLM2-135M	0.14B	-82.61 [-93.89, -71.79]	-1.23 [-1.32, -1.14]	84.00 [73.16, 95.33]	-1.39 [-1.88, -0.91]	1.56 [0.93, 2.23]
SmolLM2-360M	0.36B	-70.35 [-79.40, -61.71]	-1.09 [-1.19, -1.00]	71.02 [62.08, 80.28]	-0.67 [-1.16, -0.19]	-0.80 [-1.23, -0.40]
Qwen2.5-0.5B	0.49B	-67.68 [-76.71, -59.15]	-1.21 [-1.36, -1.08]	69.50 [61.01, 78.50]	-1.82 [-2.48, -1.19]	-0.51 [-0.95, -0.08]

Table 4: Probe results on CodeRepair (MBPP program repair). All quantities are in nats and are differences in the summed log-probability of the *same* action string under two contexts. $G > 0$ means the harness’s failure record made the failed call less likely; $G < 0$ is feedback inversion. Cells give the mean over items with a 95% cluster bootstrap interval (clustered on task); bold marks intervals excluding zero.

Model	Params	p_{rep} before	p_{rep} after	Δp_{rep}	greedy before	greedy after	greedy abs.
SmolLM2-135M	0.14B	0.142 [0.086, 0.201]	0.875 [0.840, 0.910]	0.733 [0.677, 0.789]	0.000 [0.000, 0.000]	0.430 [0.340, 0.519]	0.000 [0.000, 0.000]
SmolLM2-360M	0.36B	0.117 [0.067, 0.170]	0.753 [0.684, 0.820]	0.637 [0.568, 0.708]	0.000 [0.000, 0.000]	0.410 [0.306, 0.521]	0.020 [0.000, 0.050]
Qwen2.5-0.5B	0.49B	0.013 [0.004, 0.028]	0.602 [0.528, 0.681]	0.589 [0.516, 0.666]	0.000 [0.000, 0.000]	0.030 [0.000, 0.067]	0.000 [0.000, 0.000]
Qwen3-0.6B	0.60B	0.006 [0.000, 0.014]	0.296 [0.196, 0.399]	0.290 [0.193, 0.393]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Llama-3.2-1B	1.24B	0.009 [0.000, 0.025]	0.246 [0.136, 0.363]	0.237 [0.131, 0.348]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
SmolLM2-1.7B	1.71B	0.050 [0.021, 0.085]	0.476 [0.393, 0.570]	0.426 [0.353, 0.506]	0.000 [0.000, 0.000]	0.270 [0.174, 0.381]	0.000 [0.000, 0.000]

Table 5: Repetition probabilities on toolshed. p_{rep} is the normalised probability of the failed call among the scored candidate actions; *greedy* is the fraction of items on which greedy decoding reproduces the failed call exactly.

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199 A Full results tables

200 The main text reports the headline quantities. This appendix carries the material the six-page limit

201 excludes: the second environment, the repetition readouts, the robustness variants, the per-error-

202 family breakdown, the probe conditions and perturbation operators in full, the verbatim prompts,

203 and example trajectories under every harness.

204 **Program repair.** The same measurement on MBPP, where the observations are the Python in-

205 terpreter’s own output. Actions here are whole functions, an order of magnitude longer than tool

206 calls, so raw nats are not comparable with ToolShed; divided by action length they are, and both

207 environments land at close to one nat per action token.

208 **Repetition readouts.** p_{rep} is the normalised probability of the failed call over the scored candidate

209 set; *greedy* is the fraction of items on which greedy decoding reproduces the failed call exactly.

Model	Params	p_{rep} before	p_{rep} after	Δp_{rep}	greedy before	greedy after	greedy abs.
SmolLM2-135M	0.14B	0.111 [0.053, 0.180]	0.951 [0.917, 0.978]	0.840 [0.770, 0.901]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
SmolLM2-360M	0.36B	0.089 [0.032, 0.156]	0.856 [0.795, 0.910]	0.767 [0.689, 0.840]	0.000 [0.000, 0.000]	0.100 [0.033, 0.183]	0.000 [0.000, 0.000]
Qwen2.5-0.5B	0.49B	0.075 [0.023, 0.138]	0.833 [0.766, 0.893]	0.758 [0.680, 0.831]	0.000 [0.000, 0.000]	0.017 [0.000, 0.050]	0.000 [0.000, 0.000]

Table 6: Repetition probabilities on coderepair. p_{rep} is the normalised probability of the failed call among the scored candidate actions; *greedy* is the fraction of items on which greedy decoding reproduces the failed call exactly.

Model	Params	frac. $G < 0$	Δmargin	repair	copy [✓]	polarity
SmolLM2-135M	0.14B	1.00 [1.00, 1.00]	-12.58 [-14.34, -10.90]	11.34 [10.21, 12.40]	24.46 [22.57, 26.51]	-0.54 [-0.93, -0.15]
SmolLM2-360M	0.36B	1.00 [1.00, 1.00]	-12.23 [-14.11, -10.51]	6.39 [4.86, 8.02]	17.80 [15.59, 20.22]	0.82 [0.20, 1.49]
Qwen2.5-0.5B	0.49B	0.90 [0.84, 0.95]	-15.16 [-17.71, -12.79]	-1.25 [-2.46, 0.03]	11.01 [8.77, 13.37]	2.90 [1.73, 4.08]
Qwen3-0.6B	0.60B	0.94 [0.88, 0.99]	-17.54 [-20.69, -14.61]	3.96 [2.25, 5.75]	21.03 [17.85, 24.38]	0.47 [-0.41, 1.28]
Llama-3.2-1B	1.24B	1.00 [1.00, 1.00]	-9.60 [-12.31, -7.36]	3.08 [1.20, 4.96]	12.51 [9.39, 15.93]	0.17 [-1.07, 1.30]
SmolLM2-1.7B	1.71B	1.00 [1.00, 1.00]	-11.61 [-13.39, -9.97]	2.30 [1.09, 3.59]	12.92 [11.48, 14.48]	0.99 [0.59, 1.41]

Table 7: Robustness and secondary quantities. *frac. $G < 0$* is the assumption-light item-level sign statistic; Δmargin is the change in the gold-minus-failed log-probability gap; *repair* is the change in the correct action’s log-probability. *copy[✓]* repeats the surface-form term against the counterfactual *success* observation instead of the valence-free one, in case the latter reads as mildly positive; the matching semantic term referenced the same way is algebraically the polarity contrast, so it appears once, in the last column.

Robustness. The assumption-light item-level sign statistic, the change in the gold-minus-failed margin, and the decomposition referenced to the counterfactual *success* observation rather than the valence-free one. The two references bracket any reasonable notion of neutrality.

B Per-error-family results

The effect is present in every error family. A result concentrated in one would have suggested that the wording of a single error message, rather than the mechanism we describe, was responsible.

Error family	SmolLM2-135M	SmolLM2-360M	Qwen2.5-0.5B	Qwen3-0.6B	Llama-3.2-1B	SmolLM2-1.7B
permission_error	-44.4 [-47.0, -42.2]	-43.0 [-44.9, -41.3]	-37.9 [-42.4, -33.9]	-51.9 [-52.9, -51.1]	-	-30.1 [-33.1, -27.4]
contact_not_found	-34.4 [-36.4, -31.3]	-21.9 [-23.0, -20.5]	-27.8 [-30.9, -26.1]	-	-	-19.0 [-21.3, -17.5]
format_error	-22.9 [-24.5, -21.1]	-18.4 [-20.1, -16.9]	-15.1 [-17.8, -12.4]	-23.9 [-27.5, -19.9]	-14.7 [-18.2, -11.6]	-13.2 [-14.5, -12.1]
file_not_found	-	-	-	-	-	-16.9 [-18.7, -14.1]
event_not_found	-18.7 [-19.2, -18.4]	-13.0 [-13.7, -12.3]	-11.7 [-12.4, -11.2]	-	-	-
missing_argument	-17.3 [-20.2, -13.6]	-15.4 [-18.6, -11.5]	-10.4 [-12.0, -8.3]	-22.7 [-24.7, -19.3]	-9.0 [-15.0, -1.1]	-9.8 [-12.1, -7.2]
unknown_tool	-18.6 [-22.4, -15.4]	-13.5 [-18.9, -9.0]	-9.8 [-12.3, -7.4]	-16.1 [-19.9, -13.2]	-11.0 [-15.4, -7.5]	-12.2 [-14.6, -9.9]
attendee_not_found	-	-	-	-	-	-12.3 [-12.9, -11.1]
unexpected_argument	-19.0 [-22.9, -14.8]	-12.3 [-16.4, -8.8]	-5.9 [-10.2, -1.1]	-9.1 [-13.4, -5.5]	-8.6 [-11.7, -5.8]	-10.1 [-11.4, -8.8]
type_error	-18.9 [-19.6, -18.2]	-11.2 [-12.6, -9.8]	-0.7 [-3.2, 1.8]	-6.3 [-11.8, -1.3]	-3.2 [-4.0, -2.3]	-7.7 [-10.2, -5.3]

Table 8: Corrective gain (nats) by error family on ToolShed, sorted by pooled mean. Rows are the error the failing call provoked; a broadly uniform column argues against any explanation resting on the wording of one error type.

C Probe conditions and perturbation operators

Every condition shares one item’s system prompt, goal and successful prefix byte for byte; only the observation the harness records differs.

Condition	What the transcript contains
pre	context before the attempt (baseline)
fail	attempt + error observation (standard ReAct harness)
succ	attempt + counterfactual success observation
neut	attempt + valence-free observation
neut_pad	attempt + valence-free observation, length-matched to 'fail'
abstract	harness-generated failure description, no verbatim call
fail_echo	as 'fail', but the error quotes the failed call verbatim
fail_terse	as 'fail', with the shortest error rendering
fail_verbose	as 'fail', with the longest error rendering
fail_instr	as 'fail', plus an explicit do-not-repeat instruction
fail_k2	the same call fails twice
fail_k3	the same call fails three times
fail_early	as 'fail', but the failure precedes the successful steps
fail_other	a DIFFERENT wrong action failed; the scored one was never written
abstract_min	bare failure marker, no diagnosis and no verbatim call

Table 9: Every probe condition. All conditions for one item share the same system prompt, goal and successful prefix, byte for byte.

Operator	Reference call	Perturbed call	Error family
arg_typo	find_contact(name='Dana Bergmann')	find_contact(contact_name='Dana Bergmann')	unexpected_argument
date_format	list_events(date='2026-09-19')	list_events(date='September 19, 2026')	format_error
drop_arg	send_message(to='dana.bergmann@corp.example', subject='agenda', body='roadmap, hiring, i...')	send_message(to='dana.bergmann@corp.example', subject='agenda')	missing_argument
hallucinated_entity	find_contact(name='Dana Bergmann')	find_contact(name='Alex Ramirez')	contact_not_found
name_for_email	send_message(to='dana.bergmann@corp.example', subject='agenda', body='roadmap, hiring, i...')	send_message(to='Dana Bergmann', subject='agenda', body='roadmap, hiring, incident review...')	format_error
readonly_target	write_file(path='reports/roster.md', content='Elena Okafor, Karim Okafor')	write_file(path='system/config.ini', content='Elena Okafor, Karim Okafor')	permission_error
tool_typo	find_contact(name='Dana Bergmann')	find_contacts(name='Dana Bergmann')	unknown_tool
type_swap	create_event(title='1:1', date='2026-10-13', time='09:30', attendees=['karim.silva@corp...'])	create_event(title='1:1', date='2026-10-13', time='09:30', attendees='karim.silva@corp.e...')	type_error

Table 10: Perturbation operators used to build failing actions in ToolShed, with one instantiated example each.

219 D Prompts

220 Emitted from the code rather than transcribed, so they cannot drift. Two typesetting conventions
221 apply and nothing else has been altered: a line too long for the page is broken and the broken line
222 ends in a backslash, and the few non-ASCII characters the prompts contain are transliterated, since
223 the typewriter font has no glyph for them.

224 ToolShed system prompt (probe)

225 You are a tool-using assistant working inside a small office workspace. Solve the user's request by \
226 calling tools one at a time.
227
228 Rules:
229 - Reply with exactly one tool call and nothing else.

```

230 - Use the form tool_name(arg=value), with keyword arguments.
231 - Dates are YYYY-MM-DD and times are HH:MM.
232 - When the request is fully satisfied, reply with finish().
233
234 Available tools:
235 - list_files(folder: str) -> list[str]
236     List the files inside a folder.
237     folder: folder name, e.g. 'notes'
238 - read_file(path: str) -> str
239     Return the contents of a file.
240     path: full path, e.g. 'notes/todo.md'
241 - write_file(path: str, content: str) -> str
242     Create or overwrite a file.
243     path: full path
244     content: text to store
245 - delete_file(path: str) -> str
246     Delete a file.
247     path: full path
248 - find_contact(name: str) -> dict
249     Look up one contact by full name.
250     name: exact full name
251 - list_contacts(team: str) -> list[dict]
252     List the members of a team.
253     team: team name, e.g. 'design'
254 - list_events(date: str) -> list[dict]
255     List calendar events on a date.
256     date: date as YYYY-MM-DD
257 - create_event(title: str, date: str, time: str, attendees: list[str]) -> str
258     Add a calendar event.
259     title: event title
260     date: date as YYYY-MM-DD
261     time: start time as HH:MM
262     attendees: list of attendee emails
263 - cancel_event(event_id: str) -> str
264     Cancel a calendar event by id.
265     event_id: event id, e.g. 'evt-4'
266 - send_message(to: str, subject: str, body: str) -> str
267     Send a message to one recipient email address.
268     to: recipient email address
269     subject: subject line
270     body: message body
271 - set_reminder(text: str, date: str) -> str
272     Store a reminder for a date.
273     text: reminder text
274     date: date as YYYY-MM-DD
275 - convert_units(value: float, from_unit: str, to_unit: str) -> float
276     Convert a value between two units.
277     value: numeric value
278     from_unit: source unit
279     to_unit: target unit

```

280 Format demonstration appended in the agent study

```

281
282 A valid reply looks exactly like this, with no prefix or explanation:
283 list_files(folder='notes')

```

284 CodeRepair system prompt

```

285 You are a Python programmer. The user gives a task and the tests that will be run against your code.
286
287 Rules:
288 - Reply with one Python function definition and nothing else.
289 - Do not include the tests, comments, explanations or markdown fences.
290 - Use exactly the function name that appears in the tests.

```

291 Do-not-repeat instruction

```

292 Important: some of your earlier tool calls failed. Do not repeat a call that has already failed; change \
293 the call before trying again.

```

294 Failure abstraction (example)

```

295 [attempt 1 failed: find_contact --- an argument name was not recognised; that call is not repeatable]

```

296 Error-family glosses used by the abstraction

```

297 arity_error            -> wrong number of arguments
298 attendee_not_found    -> no such attendee
299 contact_not_found      -> no such contact
300 event_not_found        -> no such event
301 file_not_found         -> no such file
302 folder_not_found       -> no such folder
303 format_error           -> an argument was badly formatted
304 missing_argument       -> a required argument was missing
305 parse_error            -> the call could not be parsed
306 permission_error       -> the target was read-only
307 recipient_not_found    -> no such recipient
308 runtime_error          -> the code raised an exception
309 state_error            -> the target was already in the requested state
310 syntax_error           -> the code did not compile
311 team_not_found         -> no such team
312 test_failure           -> the code did not pass the tests
313 type_error             -> an argument had the wrong type
314 unexpected_argument     -> an argument name was not recognised
315 unknown_tool           -> no such tool
316 unsupported_conversion -> that conversion is not supported

```

317 E Example trajectories

318 One task under every harness, chosen by rule: the task with the most exact repeats under the standard
319 harness. It is a single worst-case task, so it illustrates rather than demonstrates, and it complicates
320 the aggregates in two ways we would rather show than hide. The abstraction reduces repetition
321 across the task set but does nothing on this item, and the instruction does nothing across the task set
322 but changes the trajectory here.

323 qwen2.5-0.5b, harness=verbatim

```

324 goal: agent-0011 solved=False
325 1. [ok ] convert_units(value=88, from_unit='km', to_unit='mi')
326 2. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
327 3. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
328 4. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
329 5. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
330 6. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))

```

331 qwen2.5-0.5b, harness=verbatim-instr

```

332 goal: agent-0011 solved=False
333 1. [ok ] convert_units(value=88, from_unit='km', to_unit='mi')
334 2. [ERR] read_file(path='observations/54.6806.txt')
335 3. [ERR] delete_file(path='observations/54.6806.txt')
336 4. [ERR] list_files(folder='observations')
337 5. [ok ] list_files(folder='drafts')
338 6. [ok ] list_files(folder='drafts')

```

339 qwen2.5-0.5b, harness=drop

```

340 goal: agent-0011 solved=False
341 1. [ok ] convert_units(value=88, from_unit='km', to_unit='mi')
342 2. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
343 3. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
344 4. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
345 5. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
346 6. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))

```

347 qwen2.5-0.5b, harness=abstract

```

348 goal: agent-0011 solved=False
349 1. [ok ] convert_units(value=88, from_unit='km', to_unit='mi')
350 2. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
351 3. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
352 4. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
353 5. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
354 6. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))

```

355 qwen2.5-0.5b, harness=verbatim-ban

```

356 goal: agent-0011 solved=False
357 1. [ok ] convert_units(value=88, from_unit='km', to_unit='mi')
358 2. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
359 3. [ERR] write_file(path='reports/conversion.md', content=str(54.6806), error='write_file: argument value ~

```

Harness	Qwen2.5-0.5B			
	gen	prompt	ctx/step	steps
verbatim (standard)	67	2764	633	4
+ “do not repeat”	47	2210	646	3
abstract	64	2602	616	4
drop	69	2587	586	4
verbatim + ban	69	2764	633	4
abstract + ban	66	2602	616	4

Table 11: Cost per rollout. *gen* and *prompt* are tokens summed over a rollout’s steps, so a harness that takes more steps accumulates more of both; *ctx/step* divides the prompt total by the number of steps and is the fair comparison of context size. The decoder ban adds no tokens at all. The abstraction produces a slightly shorter context per step while taking more steps, because it does not terminate early on a loop. Wall-clock is omitted: these runs shared a CPU with other jobs, so it measures scheduling rather than the harness.

```

360 4. [ERR] write_file(path='reports/conversion.md', content=str(54.6806), error='write_file: SyntaxError: co-
361 5. [ERR] write_file(path='reports/conversion.md', content=str(54.6806), error='write_file: SyntaxError: co-
362 6. [ERR] write_file(path='reports/conversion.md', content=str(54.6806), error='write_file: argument value -

```

363 qwen2.5-0.5b, harness=abstract-ban

```

364 goal: agent-0011 solved=False
365 1. [ok ] convert_units(value=88, from_unit='km', to_unit='mi')
366 2. [ERR] write_file(path='reports/conversion.md', content=str(54.6806))
367 3. [ERR] write_file(path='reports/conversion.md', content=str(54.6806).replace('.', ','))
368 4. [ERR] write_file(path='reports/conversion.md', content=str(54.6806).replace('.', ',').replace('-', ','))
369 5. [ERR] write_file(path='reports/conversion.md', content=str(54.6806).replace('.', ',').replace('-', ','))
370 6. [ERR] write_file(path='reports/conversion.md', content=str(54.6806).replace('.', ',').replace('-', ','))

```

371 F Cost

372 Everything was produced on one laptop CPU (Intel Core i7-6820HQ CPU @ 2.70GHz, four physical
373 cores, 17 GB RAM) with no GPU, in float32. The probe is built out of teacher-forced scoring rather
374 than generation because on a CPU a single decoding step costs about as much as thirty scored
375 positions; prefix reuse across conditions removes 87% of the token positions a naive implementation
376 would push through the network. Both recommended interventions are close to free: the abstraction
377 shortens the context and costs no extra model call, and the ban adds one pass over a short list per
378 decoding step and no generated tokens.

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